Mohamed Ouais Hamdi

Engineering Student · Software Engineering, passionate about AI & Machine Learning
ouaishamdi8@gmail.com | Paris, France | medouaishamdi.github.io | github.com/medouaishamdi | linkedin.com/in/medouaishamdi

First-year engineering student at ESIEE-IT (Paris) with a software-engineering background from IIT Sfax. I am passionate about AI and machine learning, which I learn on my own by building complete projects and testing them on data they have never seen, and I plan to choose the AI specialization next year. Looking for my end-of-studies internship (PFE) in AI, data or software engineering.

EDUCATION

Engineering Cycle, 1st year (international exchange) | ESIEE-IT, Paris *2026 – 2027*
- Common engineering core; I plan to choose the AI specialization next year

B.Sc. in Computer Science (Software Engineering & Information Systems) | IIT, Sfax *2024 – 2027*
- Year 1 average: 15.70/20 · Year 2 average: 17.25/20 · Courses: object-oriented programming (Java, C#), .NET and ASP.NET Core, databases, web development, software testing, compilation techniques

Baccalaureate in Mathematics | Lycée Ibn Rachik, Sfax *2024*

EXPERIENCE

Full-Stack Software Engineering Intern | DualCom Consulting, Beirut (remote) *Jul – Aug 2026*
- Built Restaurant Pro, a restaurant ordering web app: REST API in Node.js, Express and TypeScript on MongoDB, JWT authentication with admin and client roles
- React + TypeScript interface with cart, order tracking and an admin dashboard

PROJECTS

Ouais FC (portfolio) | *TypeScript, Three.js, cannon-es, Web Audio* play it: medouaishamdi.github.io
- My portfolio as a 3D football game in the browser: each goal opens a project, with physics, synthesized sound, EN/FR, and a real ML model running in the visitor's browser

HistoLens | *PyTorch, ResNet-18, ONNX Runtime, FastAPI* live demo · code
- Colorectal tissue classification (9 types), trained on one hospital and tested on 7,180 images from another: 95.7% accuracy; calibrated confidence and a referral threshold

WattCast | *LightGBM, pandas, FastAPI* live demo · code
- Day-ahead forecast of France's hourly electricity demand, backtested over 7.5 years: 2.18% MAPE vs 2.24% for RTE's published forecast

ThreatScope | *LightGBM, autoencoder, SHAP, FastAPI* live demo · code
- Hybrid intrusion detection catching 56.8% of 17 attack types never seen in training (33.3% supervised alone) at 3.2% false alarms, each alert explained

SentinelVision | *YOLO11, ONNX Runtime, Kalman filter* live demo · code
- Real-time detection and tracking of people and vehicles with zone-intrusion and loitering alerts, at 25 FPS on a 2-core CPU

DocuMind | *Node.js, Express, Gemini API, BM25* live demo · code
- Document Q&A with streamed, cited answers; hybrid retrieval (BM25 from scratch + embeddings) evaluated on a 40-question benchmark

ACADEMIC PROJECTS (ESIEE-IT, IIT)

NITRO, remote-controlled car (group project, ESIEE-IT) | *ESP32, Arduino, Three.js* project site
- Car and controller built by a team of four: two ESP32 boards linked directly (20 commands/s), motor driven by a DRV8871 chopper, dual 7.4 V / 5 V power; project website as a 3D game with task tracking

Web applications in ASP.NET Core MVC | *C#, Entity Framework Core, ADO.NET* *2025 – 2026*
- ETraining, FastOrder and Hotel Booking: multi-entity CRUD modules, dependency injection and configuration

Unity games | *C#, Unity* *2025 – 2026*
- An endless runner with level progression and saved best scores, and a Crossy Road–style prototype built procedurally

TECHNICAL SKILLS

Programming: Python, Java, C#, C, JavaScript / TypeScript, PHP, SQL
Software & web: ASP.NET Core MVC, Entity Framework Core, Node.js / Express, React, FastAPI, MongoDB
AI / ML: PyTorch, scikit-learn, LightGBM, pandas, NumPy, ONNX Runtime, SHAP
Embedded systems: ESP32, Arduino, wireless microcontroller-to-microcontroller communication
Tools & methods: Git, Linux, Docker, pytest, Agile / Scrum, Unity
Languages: Arabic (native), English (C1), French (B2)